

Funciones medibles de variables aleatorias independientes

Corolario 1. Sea $n \in \mathbb{N}$ y $\forall i \in \mathbb{N} : m(i) \in \mathbb{N}$ y sean

- (i) $\{X_{i,j}\}_{i,j=1}^{n,m(i)}$ son variables aleatorias independientes
- (ii) $\forall i \in \mathbb{N}_n : f_i : \mathbb{R}^{m(i)} \longrightarrow \mathbb{R}$ medible.

$$\implies \{f_i(X_{i,1}, \dots, X_{i,m(i)})\}_{i=1}^n \text{ son independientes.}$$

Demostración: Vamos a dividir la demostración en pasos:

1. Definimos $\forall i \in \mathbb{N} : X_i : \Omega \longrightarrow \mathbb{R}^{m(i)}$ dada por $X_i(\omega) = (X_{i,1}(\omega), \dots, X_{i,m(i)}(\omega))$. Por el apartado `ejer-var-aleatorias-prod-suma/??` del `ejer-var-aleatorias-prod-suma/??`, X_i es un vector aleatorio. Veamos que $\{X_i\}_{i=1}^n$ son independientes. Sean $\forall i \in \mathbb{N} : E_i \in \mathcal{B}(\mathbb{R}^{m(i)})$ queremos ver que

$$\mathbb{P} \left(\bigcap_{i=1}^n X_i^{-1}(E_i) \right) = \prod_{i=1}^n \mathbb{P} (X_i^{-1}(E_i)) .$$

Basta probarlo para los E_i de la forma $E_i = E_{i,1} \times \dots \times E_{i,m(i)}$ con $E_{i,j} \in \mathcal{B}(\mathbb{R})$, ya que $\mathcal{B}(\mathbb{R}^{m(i)})$ está generada por estos conjuntos. Entonces

$$\forall i \in \mathbb{N}_n : X_i^{-1}(E_i) = \{\omega \in \Omega : \forall j \in \mathbb{N}_{m(i)} : X_{i,j}(\omega) \in E_{i,j}\} = \bigcap_{j=1}^{m(i)} X_{i,j}^{-1}(E_{i,j}).$$

$$\begin{aligned} \implies \mathbb{P} \left(\bigcap_{i=1}^n X_i^{-1}(E_i) \right) &= \mathbb{P} \left(\bigcap_{i=1}^n \bigcap_{j=1}^{m(i)} X_{i,j}^{-1}(E_{i,j}) \right) \stackrel{\text{indep}}{=} \prod_{i \in \mathbb{N}_n} \prod_{j=1}^{m(i)} \mathbb{P} (X_{i,j}^{-1}(E_{i,j})) \\ &= \prod_{i=1}^n \prod_{i \in \mathbb{N}_n} \mathbb{P} \left(\bigcap_{j=1}^{m(i)} X_{i,j}^{-1}(E_{i,j}) \right) = \prod_{i=1}^n \mathbb{P} (X_i^{-1}(E_i)) . \end{aligned}$$

2. Observamos que $\forall i \in \mathbb{N}_n : f_i(X_{i,1}, \dots, X_{i,m(i)}) = f_i \circ X_i$, entonces si consideramos

$\forall i \in \mathbb{N}_n : E_i \in \mathcal{B}(\mathbb{R}), (f_i \circ X_i)^{-1}(E_i) = X_i^{-1}(f_i^{-1}(E_i))$ donde $f_i^{-1}(E_i) \in \mathcal{B}(\mathbb{R}^{m(i)})$.

$$\implies \mathbb{P} \left(\bigcap_{i \in \mathbb{N}_n} (f_i \circ X_i)^{-1}(E_i) \right) = \mathbb{P} \left(\bigcap_{i \in \mathbb{N}_n} X_i^{-1}(f_i^{-1}(E_i)) \right) \stackrel{??}{=} \prod_{i \in \mathbb{N}_n} \mathbb{P}(X_i^{-1}(f_i^{-1}(E_i))).$$

Por tanto, $\{f_i \circ X_i\}_{i=1}^n$ son variables aleatorias independientes. ■

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